

Sciences (Physics, Chemistry, Biology)

1. States of Matter Particle Simulator

Prompt: Build a simulator showing particles in a container. Add sliders for "Temperature" and "State" (Solid, Liquid, Gas). As temperature increases, the particles should vibrate more, break bonds, and move freely, demonstrating kinetic theory.

MYP Link: Science (Criteria B & C), Concept: Change.

2. Interactive Punnett Square & Genetics Visualizer

Prompt: Create a tool where users select two parent genotypes (e.g., Bb and bb) from dropdowns. The simulator should generate the offspring probabilities, display a visual Punnett square, and show the resulting phenotype ratio with simple emoji or color-coded traits (e.g., eye color).

MYP Link: Biology, Concept: Systems.

3. Simple Pendulum Physics Lab

Prompt: Code a swinging pendulum using HTML5 Canvas. Include sliders for "Length of String," "Gravity," and "Air Resistance (Damping)." Display the real-time period of the swing and allow the user to pause and reset the simulation.

MYP Link: Physics, Concept: Relationships.

4. Predator-Prey Population Dynamics (Lotka-Volterra)

Prompt: Simulate an ecosystem with two populations (e.g., wolves and rabbits). Use sliders for "Birth Rate," "Death Rate," and "Predation Rate." Graph the populations over time on a line chart to show how they cyclically affect each other.

MYP Link: Environmental Systems, Concept: Systems.

5. Ohm's Law Circuit Visualizer

Prompt: Build a simple series circuit simulator with a battery, a switch, and a lightbulb. Add sliders for Voltage (V) and Resistance (R). Calculate and display the Current (I) in real-time, and make the lightbulb glow brighter or dimmer based on the calculated current.

MYP Link: Physics, Concept: Relationships.

6. Light Refraction & Reflection Simulator

Prompt: Create a 2D simulation where a laser beam hits a boundary between two mediums (e.g., air and water). Allow the user to change the angle of incidence and the refractive indices. Draw the reflected and refracted rays using Snell's Law.

MYP Link: Physics (Optics), Concept: Change.

Mathematics

7. Probability & Dice Distribution Grapher

Prompt: Build a simulator that rolls 2 to 5 dice simultaneously. Include a "Roll 1000 Times" button. Display the results as a dynamic bar chart (histogram) to prove how the distribution approaches a normal curve (bell curve) as the number of dice or rolls increases.

MYP Link: Math (Statistics), Concept: Logic.

8. The Monty Hall Problem Interactive

Prompt: Simulate the famous game show problem with 3 doors. Let the user play manually, but also include an "Auto-Play 10,000 times" button that tracks and displays the win percentage for "Always Switch" vs. "Never Switch" strategies.

MYP Link: Math (Probability), Concept: Logic.

9. Fractal Tree Generator

Prompt: Use recursive JavaScript functions on an HTML5 Canvas to draw a branching tree. Add sliders for "Branch Angle," "Recursion Depth," and "Length Reduction" so the user can see how small changes in rules create complex, natural-looking patterns.

MYP Link: Math / Art, Concept: Patterns.

10. Geometric Tessellation Explorer

Prompt: Create a grid-based simulator where users can select a regular polygon (triangle, square, hexagon). The code should automatically tile the shape across the canvas without gaps, demonstrating which shapes tessellate and why (interior angles adding to 360°).

MYP Link: Math (Geometry), Concept: Form.

Individuals & Societies (Humanities)

11. Ancient Trade Route Resource Manager

Prompt: Build a text-and-grid-based simulator where the user manages a caravan on the Silk Road. They must allocate limited resources (water, food, gold) to survive random events (sandstorms, bandits, market price fluctuations) and reach the destination.

MYP Link: History / Geography, Global Context: Globalization and Sustainability.

12. Population Growth & Demographic Transition Model

Prompt: Create a simulator of a fictional country. Allow the user to adjust "Birth Rate," "Death Rate," and "Immigration." Show how the population pyramid changes over 50 years and identify which stage of the Demographic Transition Model the country is in.

MYP Link: Geography, Concept: Change.

Design & Computer Science

13. Visual Sorting Algorithm Simulator

Prompt: Generate an array of random vertical bars of different heights. Add buttons to sort them using "Bubble Sort" and "Selection Sort." Animate the swapping process step-by-step so the user can visually compare the speed and mechanics of each algorithm.

MYP Link: Design (Criterion B: Planning), Concept: Systems.

14. Conway's Game of Life (Cellular Automaton)

Prompt: Build a grid where cells are either "alive" or "dead." Implement the four rules of Conway's Game of Life. Add "Start," "Pause," "Clear," and "Randomize" buttons, and let the user click cells to toggle their state before running the simulation.

MYP Link: Computer Science, Concept: Systems.

15. A or Breadth-First Search (BFS) Pathfinding Visualizer*

Prompt: Create a grid with a start node, an end node, and draggable "wall" obstacles. When the user clicks "Find Path," animate the algorithm searching the grid and highlight the shortest valid path once found.

MYP Link: Design / CS, Concept: Logic.

16. Retro Pixel Art Editor

Prompt: Build a 16x16 or 32x32 grid of clickable div elements. Include a color palette picker, an "erase" tool, and a "download as PNG" button. This teaches DOM manipulation and grid-based data structures.

MYP Link: Design (Digital Media), Concept: Communication.

Arts & Interdisciplinary

17. Interactive Color Theory Mixer

Prompt: Create a simulator with three sliders for Red, Green, and Blue (RGB). As the user adjusts them, show the resulting color in a large box. Display the HEX code, and include a "Complementary Color" generator that calculates and shows the opposite color on the wheel.

MYP Link: Visual Art / Math, Concept: Relationships.

18. Music Beat Sequencer (Step Sequencer)

Prompt: Build a simple 4x4 or 8x8 grid where rows represent different sounds (kick, snare, hi-hat, clap) and columns represent beats. Use the Web Audio API (or simple HTML5 audio elements) to play the sounds in a loop when "Play" is clicked.

MYP Link: Music / Design, Concept: Systems.

Fun & Engaging "Sandbox" Simulators

19. **2D Gravity & Collision Sandbox**

Prompt: Create a canvas where the user can click to spawn bouncing balls. Each ball should have random mass, velocity, and color. Implement basic gravity and boundary collision so they bounce off the walls and each other realistically.

MYP Link: Physics / CS, Concept: Change.

20. **Typing Speed & Accuracy Tester**

Prompt: Build a simulator that displays a random paragraph of text. When the user starts typing in an input box, track their WPM (Words Per Minute) and accuracy percentage in real-time. Highlight correctly typed letters in green and mistakes in red.

MYP Link: Language Acquisition / Design, Concept: Communication.